

GRIDBENCH · DESIGN DECISION RECORD

GridBench recommendations accepted

Design Molecule Lab · designmolecule.com

DOCUMENT CONTROL

Document ID	GBN-DDR-002	Version	0.1
Status	Draft	Date	2026-09-25
Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/gridbench		
Source	docs/decisions/0002-recommendations-accepted.md @ 5e68de8		

REVISION HISTORY

Version	Date	Author	Change
0.1	2026-09-25	Amish Chadha	Recommendations accepted by Amish (DDR-002)

0002: Recommendations accepted

- **Date:** 2026-09-25
- **Status:** accepted. Every item below that carried a recommendation is decided by Amish, 2026-09-25: go with recommendation. Items without a recommendation remain "Proposed, awaiting Amish".

■ Context

On 2026-09-25 Amish wrote, in chat: "i accept all your recommendations, go with them across all repos." Before that, GBN-DDR-001 v0.1 had adopted items A1 to A10 for TRL 3 work, open for his review, and left O1 and O2 open; the TRL 3 review note (docs/REVIEW.md , session 2026-09-25, TRL 3) added a recommendation on R7 and three suggestions. This record lists what is now decided, what changed in the repo and what stays open. TRL 4 remains on hold by Amish's instruction, and nothing here authorizes building, testing or purchasing.

■ Options considered

The options for each item are those in docs/REVIEW.md (sessions 2026-09-25, /populate and TRL 3), GBN-DDR-001 and GBN-CAL-001 v0.1. Where an item offered several options, the recommended option is the decision.

■ Decision

Table 1. Items decided by Amish, 2026-09-25: go with recommendation.

#	Item	Decision	What changed in the repo
A1	Cost options (DDR-001)	Dial indicator user-supplied; tile bought as offcut; R10 excludes the indicator	Already applied at TRL 3; status wording updated
A2	Grid standard	25 mm pitch, M6	Status wording only
A3	Worktop zones	Plywood field plus aluminum precision tile	Status wording only
A4	Insert spacing	50 mm sub-grid	Status wording only
A5	Frame	Bolted timber; welded steel as a documented variant	Status wording only
A6	Fixture bodies	Printed PETG with metal pins, screws and tile; machined option	Status wording only
A7	Stability fix	Ballast on the shelf plus optional wall anchor	Status wording only
A8	Fixture file convention	Metadata block giving grid footprint and datum	Status wording only
A9	Fixture library license	CERN-OHL-S-2.0	Status wording only
A10	Tile position	Right of the worktop	Status wording only
O2	Budget	Raise <code>budget_usd</code> from \$220 to \$250	<code>project.yaml</code> \$220 to \$250; R10 target \$250; core parts \$245.90, R10 now met on paper (\$4.10 margin)
D1	R7 on the plywood field (TRL 3 review, option a)	Flanged M6 tee nuts pressed in from the underside; screw-in inserts kept where a frame member lies below	Model: 130 tee nuts (19 mm flange, 8.0 mm hole) and 122 screw-in inserts; BOM line 5 \$25.20 to \$27.80; GBN-CAL-001 G5b to G5d; R7 met on paper at 130 positions
D2	Frame stiffness (TRL 3 review suggestion)	45 x 120 mm aprons instead of 45 x 95 mm	Model and GA; deflection under 500 N 0.49 mm to 0.37 mm; bench 40.2 kg to 41.9 kg; frame \$41.40 to \$43.50
D3	Sliding (TRL 3 review suggestion)	Rubber pads on the levelling feet	Model and GA; BOM line 2 \$2.00 to \$2.50 each; empty sliding force 82 N on hard feet, 206 N on rubber pads

Knock-on figures from GBN-CAL-001 v0.2: tipping force 107 N to 112 N empty and 176 N to 181 N ballasted; core parts \$239.20 to \$245.90.

Table 2. Items still open.

#	Item	Status
O1	First workshops and regions for co-design. No recommendation was made.	Proposed, awaiting Amish
O3	Racking stiffness: end low rails or a screwed-down shelf. The TRL 3 review named both without recommending one, and no requirement covers racking yet.	Proposed, awaiting Amish
O4	New: the 122 screw-in insert positions that sit over a frame member, where R7 is still not met (pull-out factor 0.57 to 0.92). Options: (a) rate them light duty at 0.92 N·m, giving 296 to 493 N at the part (GBN-CAL-001, G5e), and mark them on the worktop; (b) move the insert sub-grid off the frame lines; (c) relax R7 at those positions. Recommendation: (a).	Proposed, awaiting Amish

■ Consequences

- `project.yaml`: `budget_usd` \$250. No pitch or problem rewording was recommended, so none was applied. `trl` and `trl_target` stay at 3.
- GBN-PRB-001, GBN-PRC-001 and GBN-REQ-001 are revised to v0.4, GBN-CAL-001 to v0.2 and GBN-DDR-001 to v0.2. Drawing GBN-DWG-001 is at Rev P2. The concept media are re-rendered.
- Requirement status (GBN-CAL-001 v0.2): 1 not met in part (R7 at the 122 positions over the frame), 2 at risk (R3, R4), 1 not verifiable at TRL 3 (R5), 5 met on paper (R6, R8, R9, R10, R12), 3 met by design (R1, R2, R11).
- No cross-repo action arises from these decisions.
- On hold as TRL 4 work: pull tests of tee nuts and inserts, the tipping and sliding push tests with rubber pads, and any purchasing.